

Mechanical:-

CAD (fusion 360, solidworks, inventor,)

Basic mechanism,(used in robotics for the movement)

Basic knowledge on workshop(different types manufacturing processes and power tools),

Basic of rigid body dynamics (physics)

Embedded:-

Microcontrollers (Arduino, ESP32, Raspberry Pi Pico)

Sensors & Actuators (Ultrasonic, IR, IMU, Servo, DC Motors)

Communication Protocols (CAN, UART, I2C, SPI)

Signal modulation(PWM)

Embedded C / MicroPython programming

ESP- Now, Wifi and bluetooth, Mesh Topology

Software:-

Robotics Operating System (ROS 2),

Computer Vision(OPENCV),

Algorithms for Path Planning (A*, Dijkstra, RRT, etc) ,

Basics of ML:

- Supervised vs unsupervised learning

- Basic classification/regression methods

- Exploratory data analysis

- Where you would use ml vs normal deterministic methods (like where ml is actually useful)

, Simulation Environments (Gazebo, Issac Sim, MuJoCo, etc)

Again, **YOU DON'T HAVE TO LEARN THOSE SKILLS.**Just google them and try to understand why it's needed for Robotics.

!!! BEST OF LUCK !!!

